

Luke Lippstreu | Curriculum Vitae

University of Edinburgh
Higgs Centre for Theoretical Physics – Edinburgh EH9 3JZ – UK
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Academic Positions

University of Edinburgh
Postdoctoral Researcher

Edinburgh, UK
Oct 2023–present

Education

Brown University

Doctor of Philosophy (PhD)
Adviser: Marcus Spradlin

Providence, RI, USA
2018–2023

University of Cape Town

Master of Science (MSc)
Adviser: Heribert Weigert
Thesis: *Deep inelastic scattering at high energies*

Cape Town, South Africa
2016–2018

University of Cape Town

BSc (Hons), First Class

Cape Town, South Africa
2011–2015

Publications and preprints

Unitarity and the Forward Direction in Theories with Long-Range Forces
L. Lippstreu

[[arXiv:2609.16896](https://arxiv.org/abs/2609.16896)]

New Tools in the Landau Bootstrap

P. Bargiela, G. Crisanti, C. Larkin, L. Lippstreu, A. J. McLeod, M. Polackova, L. Walsh
Proceedings of Science — Loops and Legs in Quantum Field Theory 2026, article 068

[[arXiv:2607.16034](https://arxiv.org/abs/2607.16034)]

Steinmann Violation and Minimal Cuts

H. S. Hannesdottir, L. Lippstreu, A. J. McLeod, M. Polackova
Accepted for publication in JHEP

[[arXiv:2512.24010](https://arxiv.org/abs/2512.24010)]

Analytic Properties of Infrared-Finite Amplitudes in Theories with Long-Range Forces

L. Lippstreu
Accepted for publication in JHEP

[[arXiv:2505.04702](https://arxiv.org/abs/2505.04702)]

Minimal Cuts and Genealogical Constraints on Feynman Integrals

H. S. Hannesdottir, L. Lippstreu, A. J. McLeod, M. Polackova

[[arXiv:2406.05943](https://arxiv.org/abs/2406.05943)]

A Perturbation Theory for the Coulomb Phase Infrared-Divergence

L. Lippstreu
Under review at JHEP

[[arXiv:2312.08455](https://arxiv.org/abs/2312.08455)]

Landau Singularities of the 7-Point Ziggurat II

L. Lippstreu, M. Spradlin, A. Srikant, A. Volovich
JHEP 01 (2024) 069

[[arXiv:2305.17069](https://arxiv.org/abs/2305.17069)]

Landau Singularities of the 7-Point Ziggurat I

L. Lippstreu, M. Spradlin, A. Volovich
JHEP 07 (2024) 024

[[arXiv:2211.16425](https://arxiv.org/abs/2211.16425)]

Loop-Level Gluon OPEs in Celestial Holography

R. Bhardwaj, L. Lippstreu, L. Ren, M. Spradlin, A. Srikant, A. Volovich
JHEP 11 (2022) 171

[[arXiv:2208.14416](https://arxiv.org/abs/2208.14416)]

Angular Momentum of the Asymptotic Electromagnetic Field in the Classical Scattering of Charged Particles

R. Bhardwaj, L. Lippstreu

[arXiv:2208.02727]

Four-Point Correlators of Light-Ray Operators in CCFT

Y. Hu, L. Lippstreu, M. Spradlin, A. Srikant, A. Volovich

JHEP 07 (2022) 104

[arXiv:2203.04255]

Zwanziger's Pairwise Little Group on the Celestial Sphere

L. Lippstreu

JHEP 11 (2021) 051

[arXiv:2106.00084]

Weak Separation, Positivity and Extremal Yangian Invariants

L. Lippstreu, J. Mago, M. Spradlin, A. Volovich

JHEP 09 (2019) 093

[arXiv:1906.11034]

Manuscripts in preparation

New analytic constraints for the Soft Anomalous Dimension Bootstrap at three loops

E. Gardi, L. Lippstreu, A. J. McLeod, L. Walsh, Z. Zhu, E. Zunino

In preparation

The d'Alembertian Method for Wilson-Line Correlators

E. Gardi, L. Lippstreu, A. J. McLeod

In preparation

Efficient Genealogical Constraints from Sector-By-Sector Critical Point Counting

G. Crisanti, L. Lippstreu, A. J. McLeod, M. Polackova

In preparation

Conference Talks & Seminars

QCD Seminar (*scheduled*)

The d'Alembertian Method for Computing the Three-Loop Soft Anomalous Dimension

CERN Theory Department

Geneva, Switzerland

30 Nov 2026

Gravity-QFT Seminar (*scheduled*)

Unitarity and the Forward Direction in Theories with Long-Range Forces

Imperial College London

London, UK

16 Oct 2026

Scattering Amplitudes: From Positive Geometries to Analytic Structures (*workshop*)

Unitarity and the Forward Direction in Theories with Long-Range Forces

Castello dei Principi Capano

Pollica, Italy

22 Sep 2026

Amplitudes et Gravitation sur l'Yvette Seminar

Novel Properties of QFTs with Long-Range Interactions

Institut des Hautes Études Scientifiques (IHES)

Bures-sur-Yvette, France

10 Jun 2026

Prague Spring Amplitudes 2 (*workshop*)

Boxing Wilson Lines: A d'Alembertian Method for the Soft Anomalous Dimension

Institute of Experimental and Applied Physics, Czech Technical University in Prague

Prague, Czech Republic

11 May 2026

Amsterdam String Seminar

Novel Properties of QFTs with Long-Range Interactions

University of Amsterdam

Amsterdam, Netherlands

24 Mar 2026

Theory Seminar

Novel Properties of QFTs with Long-Range Interactions

Queen Mary University of London

London, UK

29 Jan 2026

Quantum Gravity and Strings Lab Seminar

Analytic Properties of Infrared-Finite Amplitudes in Theories with Long-Range Forces

University of Cape Town

Cape Town, South Africa

5 Jan 2026

QCD Meets Gravity XI (*conference*)

Analytic Properties of Infrared-Finite Amplitudes in Theories with Long-Range Forces

ICTP-SAIFR ([video](#))

São Paulo, Brazil

15 Dec 2025

HEP Seminar <i>Analytic Properties of Infrared-Finite Amplitudes in Theories with Long-Range Forces</i> McGill University	Montreal, QC, Canada 8 Sep 2025
Bootstrapping Amplitudes from Weak to Strong Coupling (workshop) <i>Automating Hierarchical Constraints for Feynman Integrals</i> CERN (video)	Geneva, Switzerland 22 May 2025
Infrared Surprises of Scattering Amplitudes (workshop) <i>Analytic Properties of Infrared-Finite Amplitudes in Theories with Long-Range Forces</i> CERN (video)	Geneva, Switzerland 16 May 2025
Amplitudes Seminar <i>Minimal Cuts and Genealogical Constraints</i> University of Bonn	Bonn, Germany 9 Jan 2025
Amplitudes Lounge <i>Minimal Cuts and Genealogical Constraints on Feynman Integrals</i> University of Padova	Online 8 Nov 2024
Higgs Amplitudes Meeting <i>Genealogical Constraints on Feynman Integrals, and Infrared-Finite Scattering Amplitudes</i> Higgs Centre for Theoretical Physics, University of Edinburgh	Edinburgh, UK 23 Jul 2024
Harvard BHI/CMSA Conference on Flat Holography <i>Loop-Level Gluon OPEs in Celestial Holography (short talk)</i> Center of Mathematical Sciences and Applications, Harvard University	Cambridge, MA, USA Jun 2022
Celestial Holography 2022 <i>Four-Point Correlators of Light-Ray Operators in CCFT (gong show)</i> Princeton University	Online Feb 2022
Institute for Advanced Study Journal Club <i>Zwanziger's Pairwise Little Group on the Celestial Sphere</i> Institute for Advanced Study	Online May 2021

Research Supervision

Laura Walsh <i>PhD Co-supervisor</i>	University of Edinburgh 2026–present
Stuart van Breda <i>MSc Project Supervisor</i> Project: <i>Hidden Simplicity in Multi-Loop Feynman Integrals</i>	University of Edinburgh Dec 2025–Aug 2026

Academic Leadership & Service

Higgs Amplitudes Seminar <i>Organizer</i>	University of Edinburgh Aug 2024–present
Amplitudes Day <i>Co-organizer</i>	University of Edinburgh 19 Feb 2026
New Frontiers in Landau Analysis (workshop) <i>Organizer</i>	University of Edinburgh 24–26 Apr 2024

Teaching Experience

Analyticity, Dispersion Relations, Positivity and All That <i>Teaching Assistant</i> Instructor: Andrew J. Tolley	9–13 Jun 2025
Teaching Assistant, Brown University	
Basic Physics A (First year)	Fall 2018, Fall 2019
Basic Physics B (First year)	Spring 2018, Spring 2019

Teaching Assistant, University of Cape Town		
Quantum Mechanics	(Fourth year)	Spring 2018
Electromagnetism	(Third year)	Spring 2018
Classical Mechanics	(Second year)	Spring 2018
Statistical Mechanics	(Fourth year)	Fall 2017
Statistical Mechanics	(Third year)	Fall 2017
Quantum Mechanics	(Second year)	Fall 2017
Physics A and B for Engineers	(First year)	2015–2018

Fellowships & Awards

Department of Physics, Brown University	Providence, RI, USA
<i>Physics Dissertation Fellowship</i>	2022–2023
Department of Physics, University of Cape Town	Cape Town, South Africa
<i>Master's Research Scholarship</i>	2016–2017
University of Cape Town	Cape Town, South Africa
<i>Joseph Stone Bursary</i>	2016–2017
University of Cape Town	Cape Town, South Africa
<i>National Research Foundation Research Scholarship</i>	2017
University of Cape Town	Cape Town, South Africa
<i>National Institute for Theoretical Physics Research Scholarship</i>	2015
University of Cape Town	Cape Town, South Africa
<i>Science Faculty Scholarship</i>	2013

Research Visits & Experience

CERN	Geneva, Switzerland
<i>Research Visit</i>	Jun 2023
Institute for Advanced Study	Princeton, NJ, USA
<i>Research Visit</i>	Mar 2023
University of Pretoria	Pretoria, South Africa
<i>Research Project</i>	Dec 2016
Project: Continuous Phase Transitions and the Renormalization Group	
Supervisor: Konstantinos Zoubos; supported by the National Institute for Theoretical Physics	

References

Marcus Spradlin <i>Professor of Physics</i> Department of Physics Brown University Box 1843, 182 Hope Street Providence, RI 02912, USA	Andrew J. McLeod <i>Royal Society University Research Fellow</i> Higgs Centre for Theoretical Physics School of Physics and Astronomy University of Edinburgh James Clerk Maxwell Building Peter Guthrie Tait Road Edinburgh EH9 3FD, UK	Einan Gardi <i>Professor</i> Director, Higgs Centre for Theoretical Physics School of Physics and Astronomy University of Edinburgh James Clerk Maxwell Building Peter Guthrie Tait Road Edinburgh EH9 3FD, UK
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